Answer key: Lesson 8, Balance and Swap

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill in the blank so both sides are the same.

Answer: 4, 5, 3, 2

(a) 5 + 3 is 8 and 4 + 4 is 8. (b) 6 + 4 is 10 and 5 + 5 is 10. (c) 7 + 3 is 10 and 3 + 7 is 10. (d) 2 + 8 is 10 and 8 + 2 is 10.

Watch for: Writing the total of the left side in the blank (8 in part a).; Thinking the equals sign means the answer comes next.

S2. Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

Answer: 1 more button (6 + 2 = 8 and 7 + 1 = 8)

The left plate has 6 + 2 = 8 buttons. The right plate has 7 buttons, so it needs 1 more to make 8.

Watch for: Drawing 8 more buttons on the right plate.; Forgetting the 7 buttons already there.

S3. Siti fills in the blank and writes 9. Is she right? Write the number that makes both sides the same. Say what Siti mixed up.

Answer: No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next.

6 + 3 is 9. The right side must also be 9. 5 and 4 more make 9, so the blank is 4. Siti wrote 9, the total of the left side, but the blank is only one of the two numbers on the right.

Watch for: Saying she is right because 9 is the answer to 6 + 3.; Not saying what she mixed up (any clear sentence about the equals sign or the right side is fine).

S4. 7 + 5 = 12. Each time, give 1 from the first number to the second number. The total stays 12. Fill in the blanks.

Answer: 6, 7, 8

(a) 6 + 6 = 12. (b) 5 + 7 = 12. (c) 4 + 8 = 12. The first number goes down by 1 and the second goes up by 1, so the total stays 12.

Watch for: Leaving the second number as 5.; Making the second number go down as well.

S5. Fill in the blanks. Look at how the first number moved.

Answer: 10, 10, 4, 9

(a) 20 went up by 20 to 40, so 30 goes down by 20 to 10. (b) 50 went up by 10 to 60, so 20 goes down by 10 to 10. (c) 34 went up by 1 to 35, so 5 goes down to 4. (d) 61 went down by 1 to 60, so 8 goes up to 9.

Watch for: Writing the total of the left side (50, 70, 39 or 69).; Moving both numbers the same way (part c: 5 + 1 = 6).

S6. Mei has 9 red pens and 6 blue pens. Ben has 10 red pens. How many blue pens must Ben have to have the same number of pens as Mei?

Answer: 5 blue pens (9 + 6 = 10 + 5)

Mei has 9 + 6 = 15 pens. Ben has 10 red pens, so he needs 5 more to make 15. Another way: his red pens are 1 more than Mei's, so his blue pens are 1 fewer: 5.

Watch for: Answering 6 (copying Mei's blue pens).; Answering 15 (the total).

S7. In this sum the 12 became 2. That is one ten less. Fill in the blank.

Answer: 45 (35 + 12 = 47 and 45 + 2 = 47)

The 12 went down by 10 to become 2, so the 35 goes up by 10 to become 45. Check: 35 + 12 = 47 and 45 + 2 = 47.

Watch for: Writing 47 (the total).; Writing 25 (taking a ten from the 35).

Think harder (page 3)

T1. Mei says 47 + 18 and 45 + 20 have the same total. Ali says 45 + 20 is bigger. Who is right? Do not add. Say how the numbers moved.

Answer: Mei. 47 went down by 2 to 45 and 18 went up by 2 to 20. The 2 moved across, so both totals are 65.

47 to 45 is 2 less. 18 to 20 is 2 more. The 2 moved from one number to the other, so the total stays the same. Both sums make 65.

Watch for: Saying 45 + 20 is bigger because 20 is a bigger number than 18.; Adding both sums and not saying how the numbers moved (correct, but check the reason).

T2. Look at each pair. Do not add yet. Write SAME if the totals match. If not, write LEFT or RIGHT for the side with the bigger total.

Answer: SAME, SAME, RIGHT

(a) 29 up 1 to 30, 14 down 1 to 13: SAME (both 43). (b) 53 down 3 to 50, 20 up 3 to 23: SAME (both 73). (c) 26 up 2 to 28 and 15 up 1 to 16: both went up, so RIGHT is bigger (41 and 44).

Watch for: Saying SAME for part c because both numbers changed.; Looking only at the biggest numbers.

T3. Make adding easier. Fill in the blank, then write the total.

Answer: 25, then the total is 75

49 went up by 1 to 50, so 26 goes down by 1 to 25. 50 + 25 = 75, so 49 + 26 = 75.

Watch for: Writing 27 (making 26 go up as well).; Writing 75 in the blank.

T4. Ben wrote 24 + 17 = 26 + 19. Is it balanced? If not, change the 19 to a number that makes it balance.

Answer: No. 24 + 17 = 41 but 26 + 19 = 45. Both numbers went up by 2. Change 19 to 15.

24 went up by 2 to 26, so 17 should go down by 2 to 15. Ben made it go up by 2 to 19, so the right side is 4 too big. Check: 24 + 17 = 41 and 26 + 15 = 41.

Watch for: Saying it is balanced because both numbers changed by 2.; Changing the 26 and not saying which number you change.

Go further (page 4)

G1. Fill in both blanks so that every sum is the same.

Answer: 27 and 17 (every sum is 47)

19 up 1 to 20, so 28 down 1 to 27. The total is 20 + 27 = 47 (check: 19 + 28 = 47). Then ___ + 30 = 47, so the second blank is 47 - 30 = 17.

Watch for: Writing 27 in both blanks.; Writing 30 or 28 as the second blank.

G2. Two blanks. Write three different pairs of numbers that fill them and balance. What do you notice about the first blank and the second blank?

Answer: Any pair where the first blank is 4 more than the second, for example 5 and 1, 6 and 2, 7 and 3 (also 4 and 0).

If the second blank is 1, the right side is 10, so the first blank is 5. If it is 2, the right side is 11, so the first blank is 6. The first blank is always 4 more than the second, because 9 is 4 more than 5.

Watch for: Writing the same number in both blanks.; Giving one pair only.

G3. Mei has 28 stickers on a page and 17 in her pocket. Ali and Ben have the same number in all. Ali has 30 on a page. How many are in his pocket?

Answer: 15 and 20 (Mei has 45 stickers; 30 + 15 = 45 and 25 + 20 = 45)

Mei has 28 + 17 = 45. Ali: 30 is 2 more than 28, so his pocket has 2 fewer than 17: 15. Ben: 25 is 3 fewer than 28, so his pocket has 3 more than 17: 20. Check: 30 + 15 = 45 and 25 + 20 = 45.

Watch for: Writing 17 for Ali's pocket.; Making Ben's pocket 14 (moving the wrong way).

Investigation (page 5): Keep it the same

Add some to one number and take the same from the other: the total stays the same (give and take). Add 3 to both: the total goes up by 6. Take 2 from both: the total goes down by 4. When both numbers change the same way, the total changes by double.

Look for: A child who tries more than one pair of numbers, says it stays the same when one goes up and the other goes down by the same amount, and notices that changing both numbers the same way changes the total by double. Any clear reason is fine.

Exit ticket (page 5)

E1. Fill in the blank.

Answer: 4

9 went up by 1 to 10, so 5 goes down by 1 to 4. Both sides make 14.

Watch for: Writing 14 (the total).; Writing 6 (making the 5 go up).

E2. Is 28 + 6 = 30 + 4 balanced? Write yes or no.

Answer: Yes

28 went up by 2 to 30 and 6 went down by 2 to 4. Both sides make 34.

Watch for: Writing no because the numbers look different.

E3. Fill in the blank. Say how the numbers moved.

Answer: 2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2.

30 went up by 10, so 12 goes down by 10 to 2. Both sides make 42.

Watch for: Writing 42 (the total).; Writing 22 (making the 12 go up).

Sneak-peek puzzles (not in the workbook)

1. Yes, the same: 10 beads each. stars.
2. 4. 9 is 1 more than 8, so the blank is 1 less than 5. (Both sides make 13.)